MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (HONOURS)

FOURTH SEMESTER EXAMINATION (2024 ADMISSION ONWARDS)

MG4DSECSC200: DATA MINING

MODEL QUESTION PAPER

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A. Very Short Answer Questions.

Answer all Questions. Each question carries 2 marks.

1. List any two data mining functionalities. [CO1][U]
2. Differentiate between OLTP and OLAP systems in one point each. [CO2][U]
3. State the two primary measures used to evaluate association rules. [CO2][U]
4. What is the primary purpose of data preprocessing in the KDD process? [CO1][U]
5. What is the role of a decision tree in classification? [CO3][U]

(5 X 2= 10 Marks)

Part B. Short Answer Questions.

Answer any 5 Questions. Each question carries 5 marks.

6. Compare and contrast the Apriori and FP-Growth algorithms. [CO2][A]
7. Explain the step-by-step process of the K-Means clustering algorithm. [CO3][A]
8. Classify the major issues in data mining into technical and application-oriented challenges. [CO1][U]
9. Construct a simple decision tree for a binary classification problem. [CO3][An]
10. Elaborate on the multidimensional data model (dimensions, facts, cubes). [CO2][A]
11. Describe the architecture of a typical data warehouse. [CO2][A]
12. Describe the steps involved in the Data Reduction process. Why is it necessary before mining? [CO1][U]

(5 X 5= 25 Marks)

Part C. Essay Questions.

Answer any 1 Questions. Each question carries 15 marks.

13. Explain constraint based association rule mining with suitable examples. [CO2][A]
14. Design a comparative framework to assess the suitability of three classification algorithms—Decision Tree, Bayesian Classification, and Support Vector Machine—for a real-world problem**.** [CO3][An]

(1 X 15 = 15 marks)